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Regular Article

Synthesis of mesoporous silica-calcium phosphate hybrid nanoparticles and their potential as efficient adsorbent for cadmium ions removal from aqueous solution

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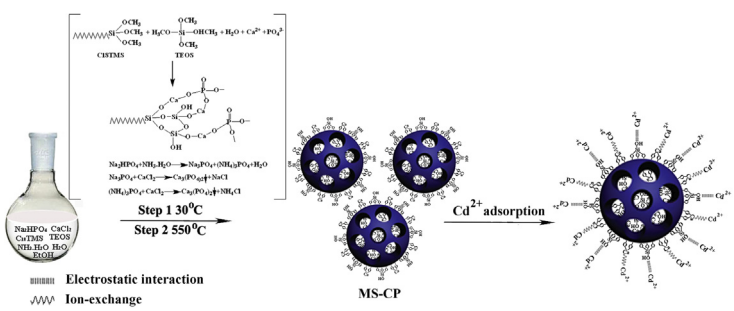
GRAPHICAL ABSTRACT



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ABSTRACT

Since adsorption and nanomaterials had been respectively found to be the most promising technique and the preferred adsorbents for heavy metal ions removal, in this study, novel mesoporous silica-calcium phosphate (MS-CP) hybrid nanoparticles were synthesized by a facile one-pot method, and subsequently assessed as adsorbent for Cd^{2+} removal from aqueous solution. MS-CP were characterized by scanning and transmission electron microscopies, etc. The influences of initial Cd^{2+} concentration, contact time, solution temperature and solution pH on removal efficiency of Cd^{2+} were investigated in detail. The results revealed that MS-CP were nanospheres of ~20 nm and presented a bimodal pore distribution (3.82 nm and 12.40 nm), a high surface area (314.56 m^2/g) and a large pore volume (1.21 cm^3/g). The Cd^{2+} removal experiments demonstrated that MS-CP had a high adsorption capacity due to electrostatic interaction between Cd^{2+} and silanol groups on MS-CP surface, as well as ion-exchange between Cd^{2+} and calcium in MS-CP. Additionally, removal efficiency of Cd^{2+} increased with increasing contact time and solution temperature, while decreased as initial Cd^{2+} concentration increased. The maximum adsorption capacity of Cd^{2+} by MS-CP was above 153 mg/L. These results suggested that the as-synthesized MS-CP could be promising adsorbent for Cd^{2+} removal from aqueous solution.

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1. Introduction

Recent advances in science and technology came with rapid industrial progress. However, such progress caused serious

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pollution in aquatic and soil systems by heavy metal ions [1–4]. The heavy metal ions pollution of aquatic environment had become a major environmental problem of worldwide concern [5]. Since heavy metal ions were not biodegradable, metabolized or decomposed [6], and some of them were toxic, carcinogenic and mutagenic [7], they imposed threat for aquatic life and ruined human health through the food chain with gradual accumulation in living organisms [6]. It was already known that after reaching and being adsorbed by living organisms, heavy metal ions could lead to multiple organ damages (e.g. hemotoxicity, nephrotoxicity, plumonotoxicity) [7], changes in cell cycle and cause carcinogenic disease [8,9]. Among the various heavy metal ions, Cd^{2+} was known as one of the most concern pollutants [10]. The major sources of Cd^{2+} were the various industrial activities such as metal refineries, electroplating, smelting, leather manufacturing, mining and photographic industry [11]. Generally, the concentration of Cd^{2+} in wastewater ranged from 3 to 55 mg/L [12]. The release and accumulation of Cd^{2+} in the aquatic environment could adversely affect water and agricultural product quality, as well as the activity of various living species [11]. Some studies convincingly proved that after entering the blood circulation, Cd^{2+} could cause severe health problems like testicular necrosis, cancer, gastrointestinal problems, renal dysfunction, teratogenic effects, and bone demineralization [13–15]. The International Agency For Research On Cancer and the United states Environmental Protection Agency had classified Cd^{2+} as a category-Icarcinogen and a group B-Icarcinogen, respectively [16]. The World Health Organization restricted that the maximum acceptable concentration of Cd^{2+} in drinking water was 0.003 mg/L [17]. Thus, from the environmental and human health perspective, the removal of Cd^{2+} from drinkable and environmental water source is a serious problem and of great important.

Until now, various treatment technologies such as chemical precipitation and oxidation, reduction, solvent extraction, ion exchange, coagulation, reverse osmosis, membrane, ultrafiltration, remediation, electrodialysis as well as adsorption had been applied for the efficient removal of heavy metal ions from aqueous solution [18–20]. Among these different removal methods, adsorption had been found to be the most promising and widely used technique because of its simplicity, high efficiency, convenience and low cost [21,22]. Several types of nanomaterials such as grapheme [23], activated carbon [24,25] and magnetic nanoparticle [26] had been investigated as efficient adsorbents to remove heavy metal ions from aqueous solution and raw wastewater. The broad application of such nanomaterials as adsorbents was connected with their specific structure, high surface area with enhanced active sites and abundant functional groups, as well as mechanical and thermal stability [7,27]. Recently, nanostructured porous silica materials as adsorbents had attracted more and more attention because of many unique properties such as high specific surface area, narrow pore distributions, tunable pore size, large pore volume, low density and good biocompatibility [22,28]. It was reported that chemical modification was usually necessary for porous silica nanoparticles to improve adsorption capacity of heavy metal ions from aquatic environment [29]. However, the chemical modification was generally complicated, time-consumed, and often processed with organosilane and organic solvents which were not friendly for environment [29].

In this study, we designed to synthesize a kind of novel mesoporous silica-calcium phosphate (MS-CP) hybrid nanoparticles as efficient adsorbent for Cd^{2+} removal by a facile one-pot method, using tetraethylorthosilicate (TEOS) as silica source, *n*-octadecyltrimethoxysilane (C18TMS) as surfactant template, calcium chloride anhydrous ($CaCl_2$) as Ca^{2+} source and disodium hydrogen phosphate dodecahydrate ($Na_2HPO_4{\cdot}12H_2O$) as PO_4^{3-} source (Scheme 1). Further, the effects of different parameters including initial Cd^{2+} concentration, contact time, solution temperature and solution pH on removing Cd^{2+} from aqueous solution were investigated.

2. Experimental procedure

2.1. Materials

Tetraethylorthosilicate (TEOS) was obtained from Tianjin Damao Chemical Co., Ltd., China. *N*-octadecyltrimethoxysilane (C18TMS) was purchased from Gelest Inc., USA. Ethanol was purchased from Tianjin Fuyu Fine Chemical Co., Ltd., China. Ammonia was purchased from Zhuzhou Quartz Glass Ltd., China. Disodium hydrogen phosphate dodecahydrate ($Na_2HPO_4{\cdot}12H_2O$) and calcium chloride anhydrous ($CaCl_2$) were obtained from Sinopharm Chemical Reagent Co., Ltd., China. Cadmium nitrate ($Cd(NO_3)_2$) was purchased from Shanghai Experimental Reagent Co., Ltd., China. All of the reagents were analytical grade and used without further purification. All solution of Cd^{2+} were prepared with deionized water, and solution pH was adjusted by 10 mM HCl or 10 mM NaOH.

2.2. Synthesis of MS-CP

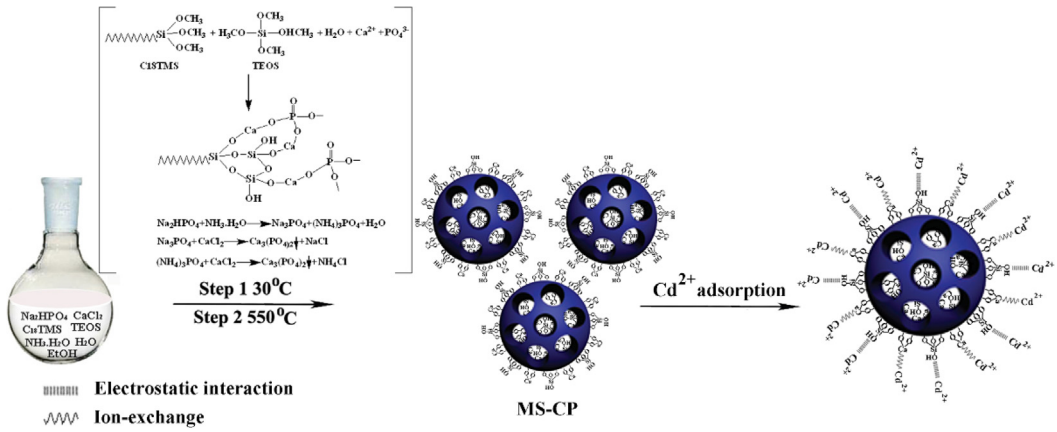
The MS-CP were synthesized according to our previous report on the preparation of mesoporous silica nanoparticles (MSN) with some modifications [30,31]. In a typical synthesis, 59 mg $Na_2HPO_4{\cdot}12H_2O$ was added into a mixture solution composed of ethanol, deionized water and ammonia under stirring. After $Na_2HPO_4{\cdot}12H_2O$ was completely dissolved, C18TMS and TEOS were added dropwise to the solution under vigorous stirring. Then, 27.88 mg $CaCl_2$ was added, and the mixture was continued to be stirred for 6 h at 30 °C. The product was collected by filtration, washed several times with deionized water and dried at 40 °C for 10–12 h. Finally, the as-synthesized product was calcined at 550 °C for 6 h to remove the template and obtain MS-CP.

2.3. Characterizations of MS-CP

Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) were taken to observe the morphology of the samples on a Hitachi S-4800 microscope at 5 kV and a JEM-2100F electron microscope operated at 200 kV, respectively. Energy dispersive X-ray spectroscopy (EDS) was performed simultaneously on JEM-2100F electron microscope to analyze surface element component. The objective lens spherical aberration correction field emission TEM was used to obtain element mapping images on a Titan G2 60–300 electron microscope at 200 kV. Wide-angle X-ray diffraction (XRD) was performed in transmission mode with a linear position sensitive detector, an anode current of 50 mA, an accelerating voltage of 32 kV and Cu X-radiation (λ = 1.5406 A°) over a scan range of 10-80° (2θ), scanning in steps of 0.004° for 23.8 s per step. The doped CP was determined by fourier transform infrared (FTIR) spectra collected on a Nicolet iS50 spectrometer. The N_2 sorption isotherms and corresponding pore size distribution were carried out to characterize the mesoporous structure of MS-CP on a NOVA-3000 analyzer at 77.3 K under a continuous adsorption condition.

2.4. Adsorption study

The adsorption behavior of the prepared MS-CP for Cd^{2+} was studied in a batch adsorption system at different solution temperatures (25, 30, 35, 40, 45, 50 °C) and different solution pH (3, 4, 5, 6, 7, 8, 9, 10). The detail experiments were carried out as following:



Scheme 1. Schematic illustration of the synthesis and Cd^{2+} adsorption process of MS-CP.

First, stock solution of Cd^{2+} was prepared in deionized water and solution pH was adjusted from 3 to 10 by 10 mM HCl or 10 mM NaOH. Then, at a certain solution temperature and a certain solution pH, 10 mg MS-CP was separately dispersed into 4 mL of Cd^{2+} solution with different concentrations (100, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600 mg/ L). Subsequently, the solution was shaken at 100 rpm for different time (0.5, 1, 2, 4, 6, 8, 12, 16, 20, 24 h). Then, the solution was centrifuged at 6000 rpm for 10 min to separate the adsorbent, and the supernatant solution was analyzed by atomic absorption spectroscopy to determine residual Cd^{2+} concentration. Finally, removal efficiency and adsorption capacity were calculated using Eq. (1) and Eq. (2), respectively:

$$\text{Removal efficiency}(\%) = \frac{(C_0 - C_e)}{C_0} \times 100\% \quad (1)$$

$$\text{Adsorption capacity} = \frac{(C_0 - C_e)}{m} \times V \quad (2)$$

Where, C_0 and C_e were the initial and equilibrium concentration (mg/L) of Cd^{2+} in the solution, respectively. V was the volume of solution (mL) and m was the weight of MS-CP (mg). To control precision, each one of the adsorption experiments was performed in triplicates and average value was reported.

2.5. Regeneration study

10 mg MS-CP was taken into a 10 mL flask containing 4 mL Cd^{2+} solution with a concentration of 200 mg/L. Then, the flask was shaken at 100 rpm for 24 h at 25 °C. The supernatant solution was separated from the adsorbent by centrifugation at 6000 rpm for 10 min, and the Cd^{2+} concentration in supernatant was measured by atomic absorption spectroscopy. The recovered MS-CP were added into 4 mL of 0.1 M of triethylenetetramine to regenerate for 6 h. The regenerated MS-CP were obtained by centrifugation, washed with deionized water several times, and dried at 70 °C for 10–12 h. Afterward, the regenerated MS-CP were employed for repeated adsorption experiment to measure the variation in removal efficiency of Cd^{2+}.

3. Results and discussions

3.1. Synthesis and characterizations of MS-CP

MS-CP were obtained by doping CP into silica framework during the synthesis process. The SEM and TEM images of MS-CP were shown in Fig. 1. It could be observed that the morphology of MS-CP was sphere with a particle size of ~20 nm.

The XRD pattern of MS-CP was given in Fig. 2a. It was clearly observed that the amorphous hump of mesoporous silica nanoparticles (MSN) in the region between about 25° and 35° was a little change after incorporation of Ca^{2+} and PO_4^{3-}. According to the XRD pattern of calcium phosphate [32], which had several peaks in the region between about 25° and 35°, the change of the amorphous hump might indicate that Ca^{2+} and PO_4^{3-} formed some calcium phosphates (CP)-like structure in MS-CP during the synthesis process.

The FTIR spectra of MS-CP was presented in Fig. 2b. The appearance of peaks at 3000–3700 and 1630 cm^{-1} might due to surface silanols and adsorbed water molecules, respectively [33–35], and the peaks at 1090, 803, and 463 cm^{-1} were assigned to Si-O-Si stretching vibration, O-Si-O bending vibration, and Si-O-Si rocking vibration, respectively [33]. Additionally, the peak at 565 cm^{-1} was attributed to Si-O-Ca asymmetric bending vibration [36]. It was notable that the peak of PO_4^{3-} groups overlapped with Si-O-Si stretching vibration at 1104 cm^{-1} [37,38]. These results indicated that the introduction of Ca^{2+} could substitute some Si within the Si-O-Si bonds, forming Si-O-Ca bonds [33].

Scanning TEM-energy dispersive X-ray elemental mapping of MS-CP was shown in Fig. 3a. It could be seen that O, Si, Ca and P signals were distributed in the particles, indicating the infiltration of CP precursors into the silica framework. The O, Si, Ca and P elements were also investigated by the EDS analysis of several different areas (Fig. 3b), the amount of O, Si, P and Ca element within MS-CP were calculated to be 56.08 wt%, 32.17 wt%, 4.02 wt% and 7.73 wt%, respectively.

The texture parameters of MS-CP were investigated by the N_2 adsorption–desorption technique. The surface area, pore size and pore volume were calculated from the desorption branch of the N_2 isotherms using BET method and BJH model, respectively. It was shown that MS-CP had a bimodal pore distribution (3.82 nm and 12.40 nm) (Fig. 4), a high surface area (314.56 m^2/g) and a large pore volume (1.21 cm^3/g).

3.2. Adsorption study

3.2.1. Effect of initial Cd^{2+} concentration

The initial metal ion concentration was an important variable to evaluate the metal ion adsorption ability of an absorbent. Therefore, in this study, the effect of initial Cd^{2+} concentration (100–600 mg/L) on removal efficiency and adsorption capacity of Cd^{2+} by MS-CP were investigated with MSN as a comparison in aqueous solution for 24 h at 25 °C (solution pH = 7, MS-CP dose = 2.5 g/L). The plots of removal efficiency and adsorption capacity of Cd^{2+} by MS-CP and MSN versus initial Cd^{2+} concentration were shown in Fig. 5. It was apparent that both removal efficiency and

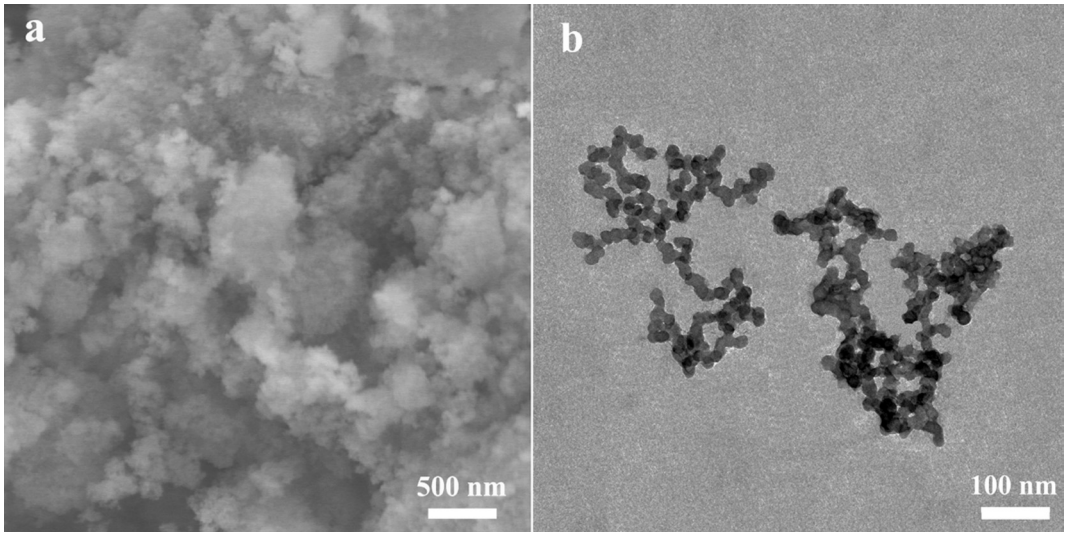


Fig. 1. The SEM (a) and TEM (b) images of MS-CP.

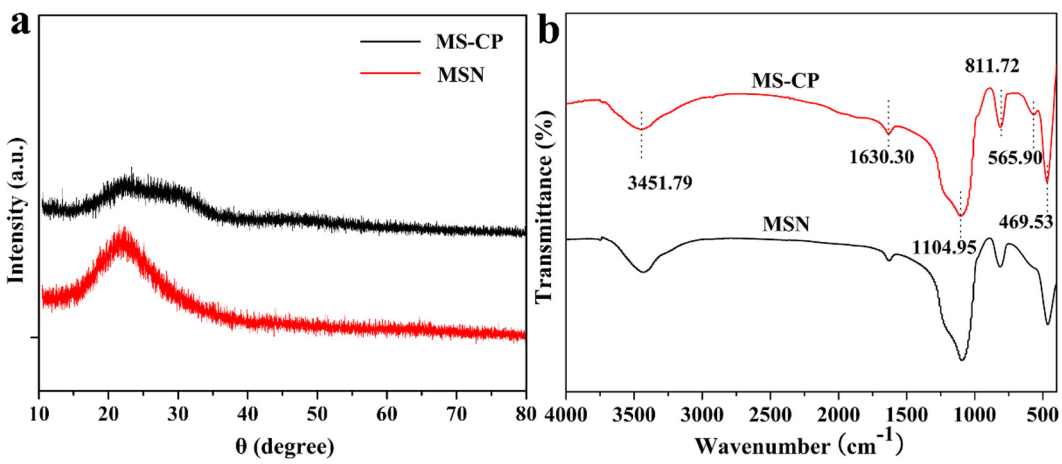


Fig. 2. The XRD patterns (a) and FTIR spectra (b) of MS-CP and MSN.

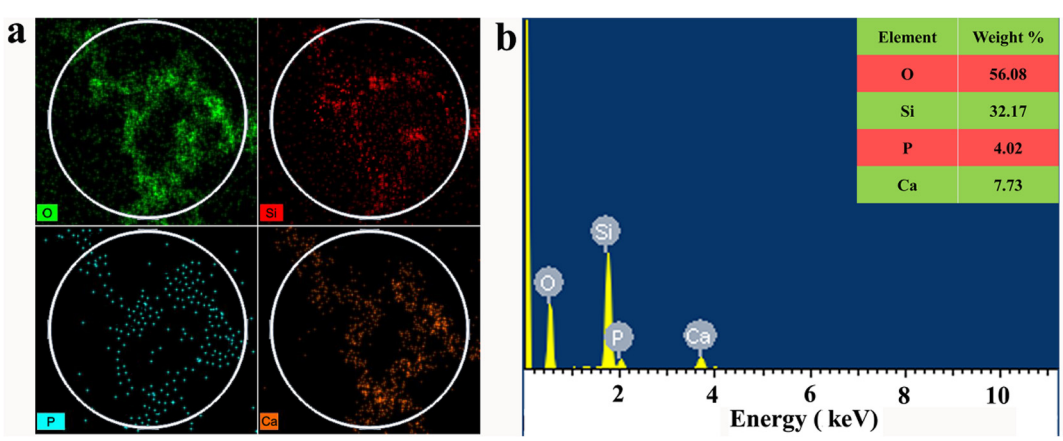


Fig. 3. Element distribution (a) and EDS spectrum (b) of MS-CP.

adsorption capacity of Cd^{2+} by MS-CP and MSN were initial Cd^{2+} concentration-dependent. However, removal efficiency and adsorption capacity of Cd^{2+} by MSN were all significantly lower than that of MS-CP. For MS-CP, removal efficiency reached 100% at the initial Cd^{2+} concentration of 100 mg/L, and decreased with the increase of initial Cd^{2+} concentration, while adsorption capacity generally increased from 40 to 153.06 mg/g as the initial Cd^{2+} concentration increased from 100 to 550 mg/L. However, adsorption capacity maintained at 153 mg/g with further increasing initial Cd^{2+} concentration to 600 mg/L, indicating reaching the

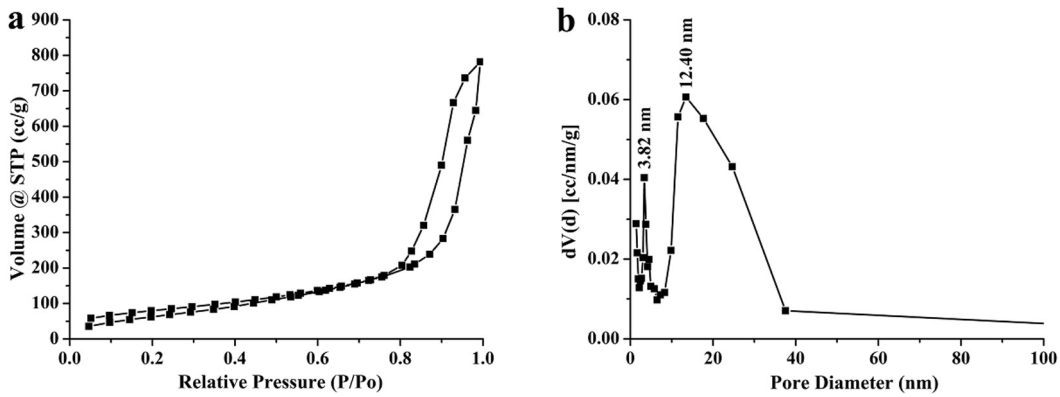


Fig. 4. N_2 adsorption-desorption isotherm (a) and pore size distribution (b) of MS-CP.

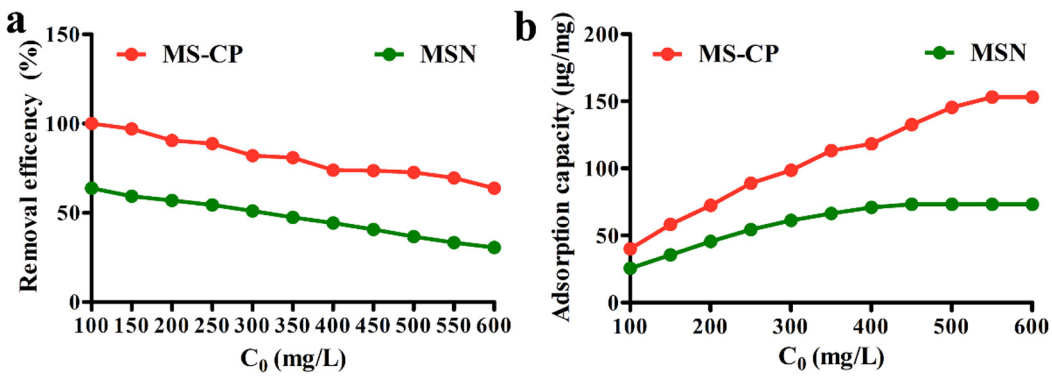


Fig. 5. Effect of initial Cd^{2+} concentration on removal efficiency (a) and adsorption capacity (b) of Cd^{2+} by MS-CP and MSN (MS-CP/MSN dose = 2.5 g/L, solution pH = 7, contact time = 24 h and solution temperature = 25 °C).

maximum adsorption capacity of Cd^{2+} by MS-CP. For MSN, the maximum adsorption capacity of Cd^{2+} was calculated to be about 73 mg/g, which was significantly lower than that of MS-CP. It was suspected that the Cd^{2+} adsorption of MSN was mainly depended on electrostatic interaction between negatively charged silanol groups on MSN surface and positively charged Cd^{2+} [39], while not only electrostatic interaction between Cd^{2+} and silanol groups on MS-CP surface [39], but also ion-exchange between Cd^{2+} and calcium in MS-CP [40,41] were the main driving forces to promote Cd^{2+} adsorption onto MS-CP (Schematic 1). Hence, the existence of calcium within silica structure played an important role in improvement of adsorption capacity of Cd^{2+} by MS-CP. At low initial Cd^{2+} concentration, sufficient adsorption sites on MS-CP surface were available for Cd^{2+} [42], thereby resulting in high removal efficiency. However, at high initial Cd^{2+} concentration, the number of Cd^{2+} was relatively higher than the number of adsorption sites on MS-CP surface, the adsorption sites were easily completely covered to reach the maximum adsorption capacity [42,43]. Therefore, an initial Cd^{2+} concentration of 100 mg/L yielded the maximum removal efficiency among all the tested concentrations.

In addition, the Cd^{2+} adsorptive property of MS-CP was compared with some recently reported adsorbents. As shown in Table 1, it could be observed that MS-CP used in our work for Cd^{2+} removal had an adsorption capacity higher or better than most of those adsorbents reported in literatures. Therefore, it could be concluded that MS-CP had a potential advantage for Cd^{2+} removal from aqueous solution.

Table 1
Comparison of Cd^{2+} adsorption capacities of MS-CP with those reported in the literatures.

Adsorbents	Adsorption capacity (mg/g)	References
NN-MCM-48	93.45	[44]
NN-MCM-41	64.93	[44]
NH_2-MCM-41/Nylon6	27.60	[45]
Melamine-based NH_2-SBA-15	71.10	[46]
NH_2-SBA-15	14.10	[46]
Polyamidoamine-SBA-15	109.00	[47]
Silica nano hollow sphere	20.82	[20]
APTS-EDTA-silica sphere	23.70	[48]
Silica HMS-SH	14.60	[49]
Silica HMS-NH_2	28.1	[49]
Nano crystallite hydroxyapatite	142.86	[50]
Poly (vinyl alcohol)/chitosan	126.06	[51]
Graphene oxide membranes	91.05	[52]
Surface functionalization of ordered mesoporous carbon	79.81	[53]
Phosphoricacid-impregnated mesoporous activated carbon	227.27	[54]
MS-CP	153.06	This work

3.2.2. Effect of contact time and solution temperature

Since the energy needed during adsorption process depended on solution temperature, and it was important that the adsorbent materials offered rapid adsorption to remove heavy metal ions from aqueous environment, it was necessary to study the effects of contact time and solution temperature on removal efficiency of Cd^{2+} by MS-CP. In this part, 10 mg MS-CP power was separately added into 4 mL solution (pH 7) containing different Cd^{2+} concentrations, and then shaken for different time (0.5, 1, 2, 4, 6, 8, 12, 16, 20, 24 h) at different solution temperatures (25 °C, 30 °C, 35 °C, 40 °C, 45 °C, 50 °C). As shown in Fig. 6, removal efficiency of Cd^{2+} by MS-CP was obviously affected by contact time and solution

temperature. Both an increase in contact time and solution temperature had positive impacts in Cd^{2+} removal. Removal efficiency significantly increased with increasing contact time and solution temperature. Within an initial Cd^{2+} concentration range of 100–350 mg/L, removal efficiency could reach 100% by increasing contact time and/ or solution temperature. However, for the initial Cd^{2+} concentration of 400 mg/L, the maximum removal efficiency was 95.67%, which might due to the limitation of the maximum adsorption capacity of Cd^{2+} by MS-CP (Fig. 5b). When keeping initial Cd^{2+} concentration constant, increasing solution temperature could effectively shorten the contact time needed for completely removing Cd^{2+} from aqueous solution. For instant, removal efficiency could reach 100% within 0.5 h when solution temperature ≥ 35 °C for initial Cd^{2+} concentration of 100 mg/L. When fixing contact time, a higher initial Cd^{2+} concentration required a higher solution temperature to promote Cd^{2+} completely adsorbed onto MS-CP. For instance, within 0.5 h, when the initial Cd^{2+} concentration increased from 150 to 200 mg/L, the corresponding solution temperature for removal efficiency of 100% increased from 35 to 50 °C. It could be explained that prolonging contact time could effectively increase the contact between Cd^{2+} and MS-CP, thereby resulting in high removal efficiency. Additionally, at high solution temperature, the high mobility of Cd^{2+} also enhanced the contact area between MS-CP and Cd^{2+}, leading to high removal efficiency [55]. These results indicated that MS-CP could rapidly adsorb Cd^{2+} with high removal efficiency, and could act as promising adsorbent for heavy metal ions removal.

3.2.3. Effect of solution pH

It was well-known that solution pH was an important factor in the adsorption process of heavy metal ions from aqueous solution because it could affect the ionizations state and the surface chemistry of the adsorbent [56,57]. Hence, in this part, the adsorption experiment was conducted in aqueous solution with a pH value ranging from 3 to 10. Fig. 7 showed the influences of different solution pH on removal efficiency of Cd^{2+} by MS-CP in an aqueous solution with an initial Cd^{2+} concentration of 200 mg/L and a MS-CP dose of 2.5 g/L for 24 h at 25 °C. The results showed that removal process was pH-dependent and could be divided into three stages: (i) Removal efficiency increased as solution pH increased from 3 to 5. At low pH, on one hand, large numbers of H_3O^+ existed in aqueous solution, competing with Cd^{2+} for adsorption sites on the surface of MS-CP, thereby resulting in low removal efficiency of Cd^{2+} by MS-CP [58]. On the other hand, due to the significant leaching of calcium from MS-CP (Fig. S1), the ion-exchange between Cd^{2+} and calcium in MS-CP decreased, leading to less Cd^{2+} adsorption on MS-CP. When solution pH increased, the number of H_3O^+ on the surface sites decreased, allowing an increasing number of Cd^{2+} to attach to the adsorption sites, thus high removal efficiency was obtained [57]. (ii) Removal efficiency almost maintained at the same level at pH 5 (99.65%) and pH 6 (99.54%). (iii) Removal efficiency increased slowly with further increasing solution pH from 7 to 10. When the alkali of the solution gradually increased, the solubility of Cd^{2+} reduced, and Cd^{2+} tended to precipitate out of the solution in the hydroxide form [59], therefore, a high density of Cd^{2+} was removed. Among all the tested solution pH, the maximum removal efficiency of Cd^{2+} by MS-CP (99.65%) was observed at pH 5.

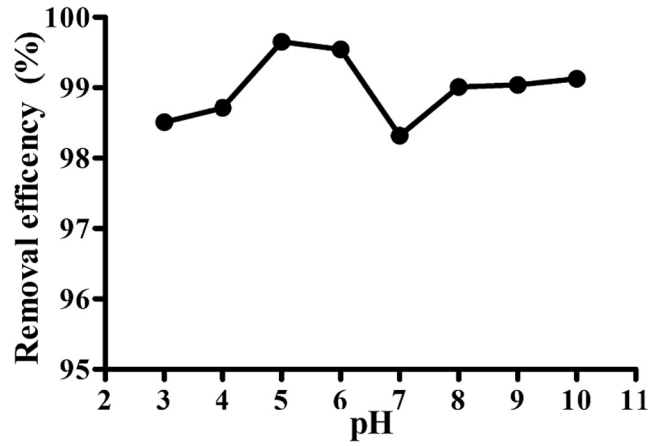


Fig. 7. Effect of solution pH on removal efficiency of Cd^{2+} by MS-CP (MS-CP dose = 2.5 g/L, solution temperature = 25 °C, initial Cd^{2+} concentration = 200 mg/L and contact time = 24 h).

3.3. Kinetic study

To determine the kinetic parameters of the Cd^{2+} adsorption process, first-order kinetic model (eq. (3)) [60], pseudo-second-order

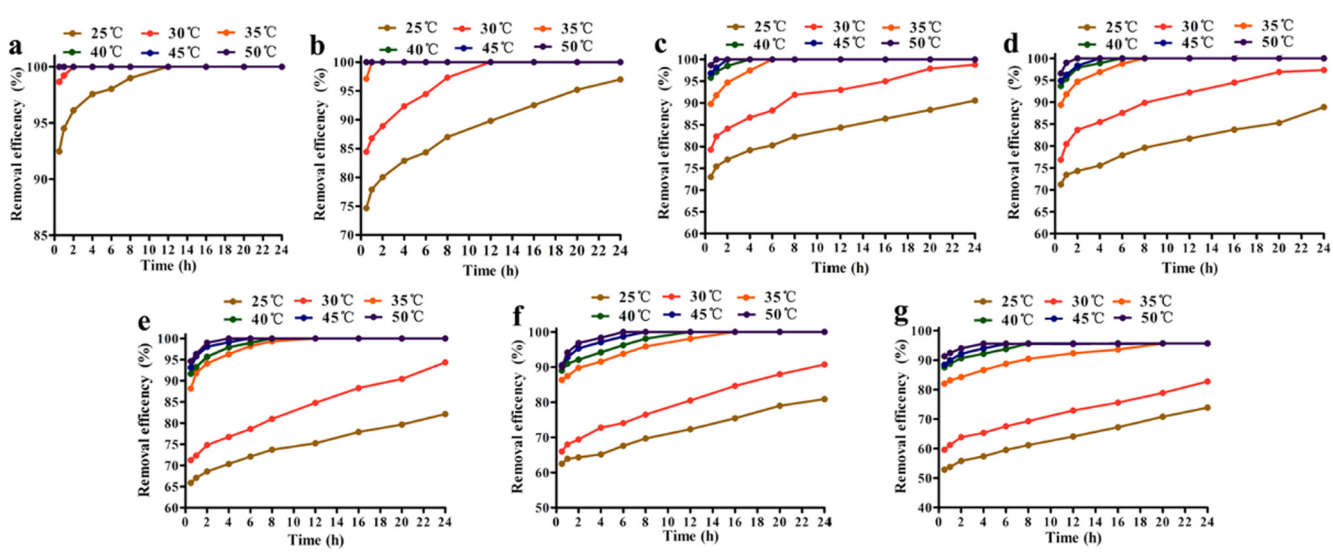


Fig. 6. Effect of contact time and solution temperature on removal efficiency of Cd^{2+} by MS-CP with initial Cd^{2+} concentration = 100 mg/L (a), 150 mg/L (b), 200 mg/L (c), 250 mg/L (d), 300 mg/L (e), 350 mg/L (f) and 400 mg/L (g) (MS-CP dose = 2.5 g/L, solution pH = 7).

model (Eq. (4)) [61], and intraparticle diffusion model (Eq. (5)) [48] were employed.

$$\ln(q_{e-}q_t) = lnq_e - k_1 t \quad (3)$$

$$t/q_t = 1/k_2 q_e^2 + (1/q_e)t \quad (4)$$

$$q_t = k_{dif} t^{0.5} + C \quad (5)$$

Where q_t, q_e, k_1, k_2 and k_{dif} were the adsorption capacity (mg g^{-1}) at timet, the equilibrium adsorption capacity (mg g^{-1}), rate constant of pseudo-first-order adsorption (min^{-1}), rate constant of pseudo-second-order adsorption (g mg^{-1} min^{-1}), and rate constant of intraparticle diffusion (mg g^{-1} $min^{-0.5}$), respectively. The values of these kinetics parameters were determined experimentally from liner plots in $(q_e - q_t)$ versus t for pseudo-first-order model (Fig. 8b), (t/q_t) versus t for pseudo-second-order model (Fig. 8c), and $q_t$versus $t^{0.5}$ for intraparticle diffusion model (Fig. 8d). The rate constant values and $q_{e,cal}$ along with correlation coefficient (R^2) were presented in Table 2. It was obvious that pseudo-second-order model demonstrated adsorption kinetics better than pseudo-first-order model with relatively higher R^2 (>0.99), indicating better adsorption process description of pseudo-second-order model compared to pseudo-first-order model.

Furthermore, intraparticle diffusion model was applied to gain insight into the diffusion mechanism of Cd^{2+} onto MS-CP. The plots of intraparticle diffusion model (Fig. 8d) exhibited the two-linear portions, implying the involvement of more than one rate controlling steps in the adsorption process. The first lines with sharper slope corresponded to the boundary layer diffusion (film diffusion) of Cd^{2+} onto MS-CP, whereas the second lines with slower slopes belonged to the intraparticle diffusion of Cd^{2+} into the interior pores of MS-CP [48]. Therefore, it could be concluded that film diffusion and intraparticle diffusion were involved in the Cd^{2+} adsorption process.

3.4. Isotherm study

The effect of concentration gradient pressure on adsorption capacity of Cd^{2+} by MS-CP was investigated at different initial Cd^{2+} concentrations ranging from 100 to 600 mg/L at 25 °C (Fig. 5), and Langmuir model (eq.(6)) [62], Freundlich model (eq. (7)) [63] and Temkin model (eq. (8)) [64] were employed to describe the Cd^{2+} adsorption process.

$$1/q_2 = (1/C_e)(1/K_L q_m) \quad (6)$$

$$\log q_e = (\log C_e/n_F) + \log K_F \quad (7)$$

$$q_e = (RT/b_T)\ln K_T + (RT/b_T)\ln C_e \quad (8)$$

In Langmuir model, q_m, C_e, and $K_L$represented the maximum adsorption capacity of MS-CP (mg g^{-1}), equilibrium Cd^{2+} concentration (mg L^{-1}), and adsorption equilibrium constant (L mg^{-1}), respectively. In Freundlich model, $K_F$and $n_F$were constants that quantified adsorption capacity ((mg g^{-1}) (L $g^{-1})^{1/n}$) and intensity, respectively. R and T indicated universal gas constant (8.314 kJ mol^{-} K^{-1}) and absolute temperature (K), respectively. In Temkin model, $b_T$and $K_T$were constants related to heat of adsorption (kJ mol^{-1}) and isotherm equilibrium binding constant (L g^{-1}), respectively.

The corresponding experimental isotherm parameters in three-type models were shown in Table 3. According to the correlation

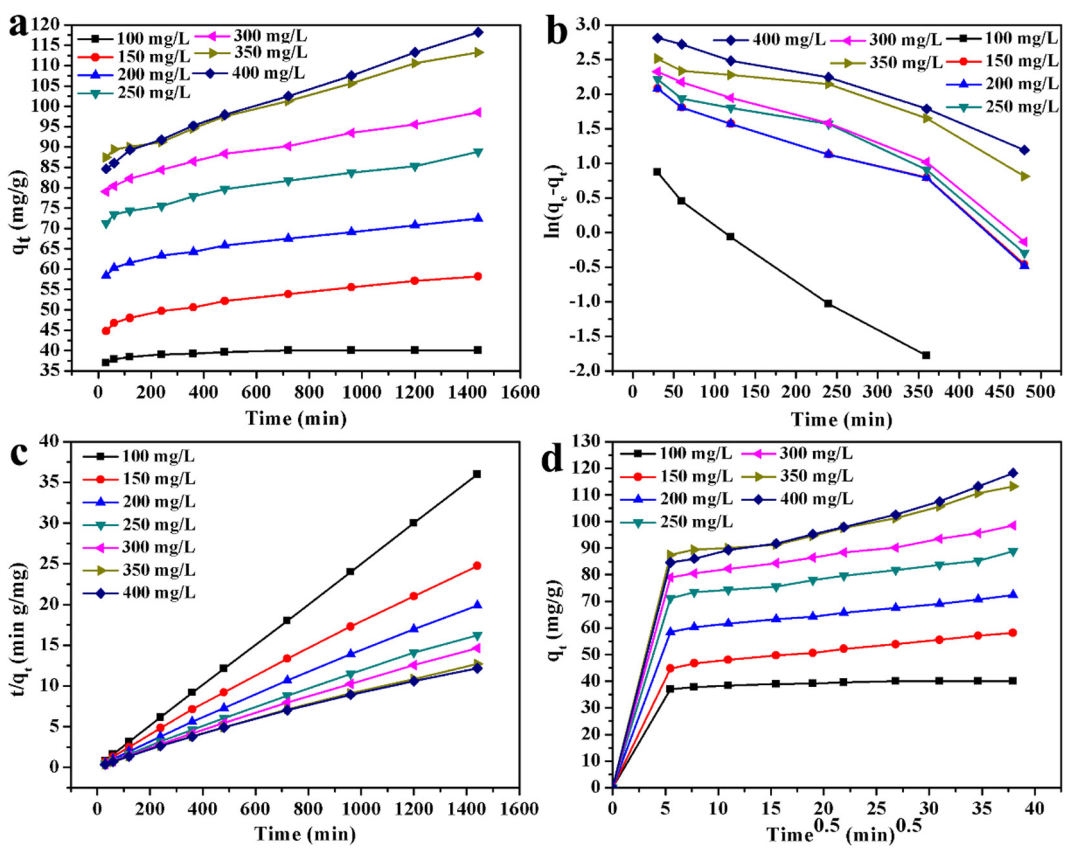


Fig. 8. Effect of contact time and initial Cd^{2+} concentration on adsorption of Cd^{2+} by MS-CP (MS-CP dose = 2.5 g/L, solution temperature = 25 °C and solution pH = 7) (a). Pseudo-first-order kinetic model (b), pseudo-second-order kinetic model (c), and intraparticle diffusion model (d) for Cd^{2+} adsorption (MS-CP dose = 2.5 g/L, solution temperature = 25 °C and solution pH = 7).

Table 2
Kinetic parameters for Cd^{2+} adsorption by MS-CP.

C_{Cd}^{2+} (mg/L)	100	150	200	250	300	350	400
Pseudo-first-order							
k_1 (min^{-1})	0.0909	0.0564	0.0651	0.0674	0.0662	0.0634	0.0161
$q_{e,cal}$ ($mg\ g^{-1}$)	39.38	52.38	66.41	80.36	89.28	99.81	101.24
R^2	0.9897	0.9514	0.9403	0.9192	0.9487	0.9023	0.9809
Pseudo-second-order							
k_2 ($g\ mg^{-1}\ min^{-1} \times 10^{-44}$)	2.37	51.38	−45.25	27.60	48.14	1.23	−2.19
$q_{e,cal}$ ($mg\ g^{-1}$)	39.11	51.68	65.34	79.16	87.89	98.11	98.65
h ($mg\ g^{-1}\ min^{-1} \times 10^{-40}$)	0.36	13.72	−19.32	17.30	37.19	1.18	−2.13
R^2	1	0.9982	0.9986	0.9979	0.998	0.9961	0.9939
Intraparticle diffusion							
k_{dif} ($mg\ g^{-1}\ min^{-1}$)	0.0853	0.3956	0.4047	0.4993	0.5771	0.7979	1.0048
C ($mg\ g^{-1}$)	37.307	43.334	56.804	68.63	75.718	81.277	77.457
R^2	0.8381	0.9949	0.9947	0.9864	0.9948	0.9666	0.9815

coefficient (R^2) values, Freundlich isotherm model ($R^2 > 0.93$) fitted experimental data better than Langmuir model and Temkin model, suggesting the reversible adsorption on multilayer heterogeneous surface and interaction between Cd^{2+} and MS-CP [45]. According to the equation of the dimensionless constant ($R_L = 1/(1 + (K_L C_O))$), the R_L values were all smaller than 0.06 (Fig. S2), owing to the positive value of K_L, strongly suggesting a favorable adsorption nature of MS-CP for Cd^{2+} [65]. In addition, the decrease of R_L with the initial Cd^{2+} concentration, indicating that the adsorption was more convenient at higher initial Cd^{2+} concentration, which was also confirmed by the n_F and K_F. The Temkin constant b_T related to the heat of adsorption ($B_T = RT/b_T$) shown in Table 3 was 1.14 kJ mol^{-1}, demonstrating the characteristic of electrostatic interaction between Cd^{2+} and MS-CP [45].

3.5. Thermodynamic study

Thermodynamic parameters were determined using the following equations:

$$\Delta G^o = -RTln(K_d) \tag{9}$$

$$\ln K_d = (\Delta S^o/R) - (\Delta H^o/RT) \tag{10}$$

Where ΔG^o, ΔS^o, ΔH^o, K_d, T and R were Gibbs free energy (kJ mol^{-1}), entropy change (J K^{-1} mol^{-1}), enthalpy change (kJ mol^{-1}), the distribution coefficient of adsorbent (mL g^{-1}), Kelvin temperature (K) and universal gas constant (8.314 J mol^{-1} K^{-1}), respectively. Thermodynamic experiments were carried out at series of Kelvin temperatures (298, 303, 308, 313, 318, 323 K), and the obtained thermodynamic parameters were given in Table 4 and Fig. S3. The positive enthalpy change (68.59 kJ) indicated the endothermic nature of the Cd^{2+} adsorption onto MS-CP [66]. In addition, the positive entropy change (292.81 J K^{-1} mol^{-1}) also suggested that the Cd^{2+} adsorption process was spontaneous, and confirmed the increased randomness of Cd^{2+} at the solid/solution interface during adsorption [66,67]. The negative ΔG^o values at all tested temperatures indicated the feasibility of the process and the spontaneous nature of the adsorption under condition applied. The ΔG^o decreased with increasing temperature suggested an increased trend in the degree of spontaneity and feasibility of Cd^{2+} adsorption [68].

Table 3
Isotherm parameters for the adsorption of Cd^{2+} onto MS-CP.

Model	Isotherm parameters	
Langmuir	q_{max} ($mg\ g^{-1}$)	125.63
	K_L ($L\ mg^{-1}$)	17.23×10^{-3}
	R^2	0.6719
Freundlich	K_F ($mg\ g^{-1}$) ($L\ g^{-1}$)$^{1/n}$	35.90
	n_F	3.70
	R^2	0.9324
Temkin	B_T	26.90
	b_T (kJ mol^{-1})	1.14
	R^2	0.9166

3.6. Regeneration study

The regeneration and reusability of MS-CP from Cd^{2+} adsorption by using 0.1 M triethylenetetramine as elution was shown in Fig. 9. It could be seen that MS-CP were efficiently eluted using triethylenetetramine and subsequently regenerated for the next cycle. However, removal efficiency of MS-CP for Cd^{2+} was slightly decreased after elution-regeneration. Removal efficiency was 90.53% for the initial adsorption, while decreased to 87.59%, 83.03 and 80.18% after the first elution-regeneration, the second elution-regeneration and the third elution-regeneration, respectively. Although the MS-CP underwent a certain irreversible change after adsorbing Cd^{2+}, removal efficiency was generally above 80%, indicating MS-CP could be potentially used repeatedly as an efficient adsorbent for Cd^{2+} removal.

4. Conclusion

In summary, MS-CP with a high surface area (314.56 m^2/g) and a large pore volume (1.21 cm^3/g), as well as a bimodal pore distribution (3.82 nm and 12.40 nm) were firstly synthesized, and subsequently assessed as efficient adsorbent for Cd^{2+} removal from aqueous solution. The Cd^{2+} removal experimental results showed that MS-CP had a high adsorption capacity due to electrostatic interaction between Cd^{2+} and silanol groups on MS-CP surface [39], as well as ion-exchange between Cd^{2+} and calcium in MS-CP [40,41]. Additionally, removal efficiency of Cd^{2+} increased with increasing contact time and solution temperature, while generally decreased as initial Cd^{2+} concentration increased. Within an initial Cd^{2+} concentration range of 100–350 mg/L, removal efficiency could reach 100% by increasing contact time and/ or solution temperature. For instance, removal efficiency reached 100% within 0.5 h when solution temperature $\geq$ 35 °C for initial Cd^{2+} concentration of 100 mg/L. When fixing contact time, a higher initial Cd^{2+} concentration required a higher solution temperature to promote Cd^{2+} completely adsorbed onto MS-CP. At the same condition, pH 5 was observed to be the most benefit for Cd^{2+} removal among all the tested pH. The maximum adsorption capacity of Cd^{2+} by MS-CP was calculated to be above 153 mg/L. In addition, compared to several other adsorbents which were recently reported in literatures [20,44–54], MS-CP showed an adsorption capacity higher or

Table 4
Thermodynamic parameters for the adsorption of Cd^{2+} onto MS-CP.

△G° (kJ mol^{-1})						△H° (kJ)	△S° (J K^{-1} mol^{-1})
298 K	303 K	308 K	313 K	318 K	323 K		
−17.42	−19.05	−23.27	−23.66	−24.02	−24.42	68.59	292.81

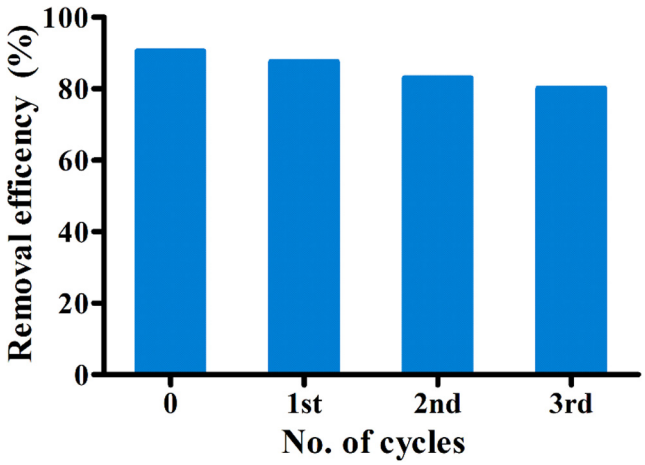


Fig. 9. Regeneration using triethylenetetramine for the multiple reuses in a series of adsorption-elution-regeneration operations (initial Cd^{2+} concentration = 200 mg/L, solution temperature = 25 °C and contact time = 24 h).

better than most of those adsorbents, indicating a considerable advantage for Cd^{2+} removal by MS-CP. Overall, these results obtained in this study demonstrated that the as-synthesized MS-CP could be promising adsorbent for the removal of heavy metal ions from aqueous solution.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.jcis.2018.04.037.

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